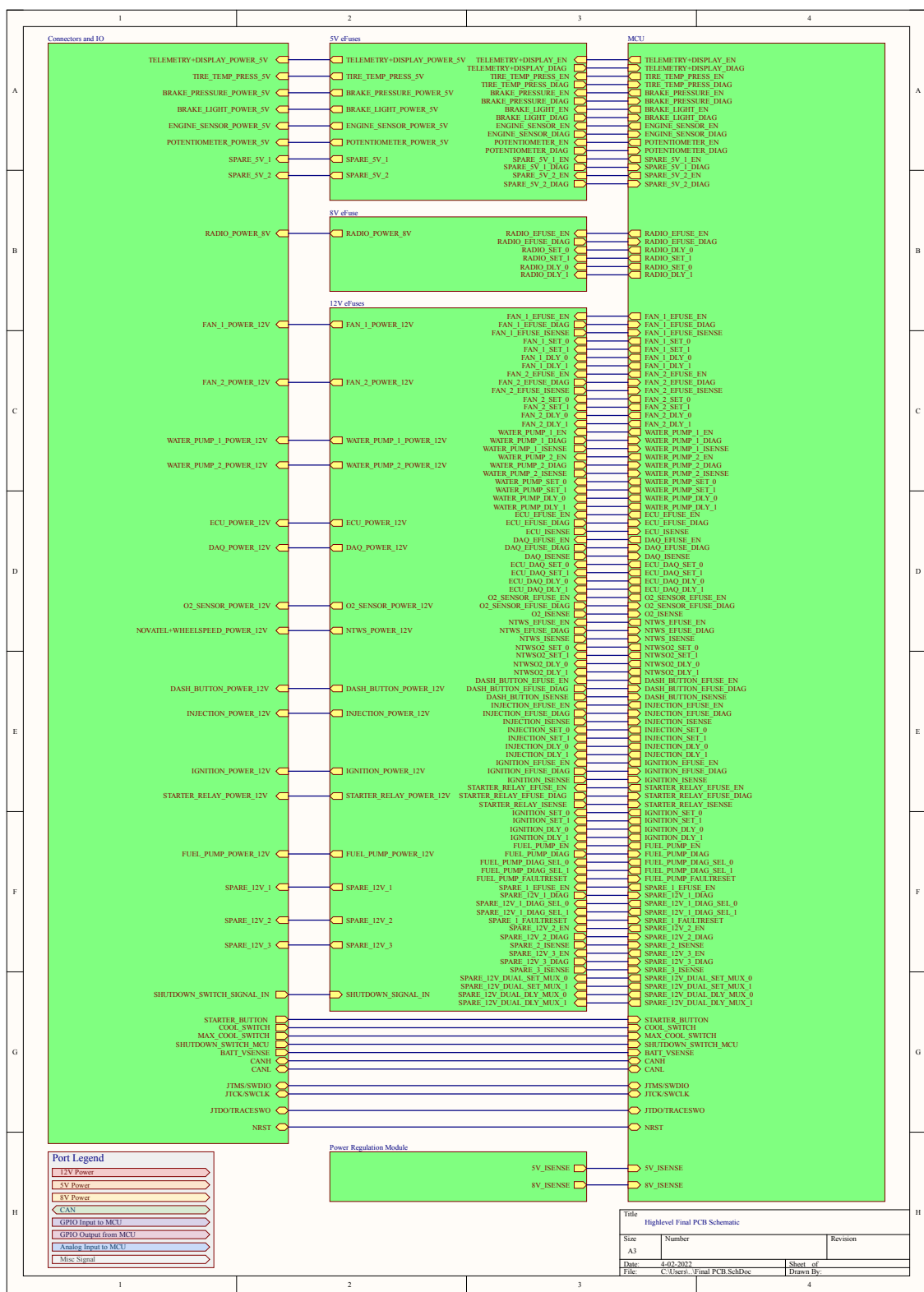
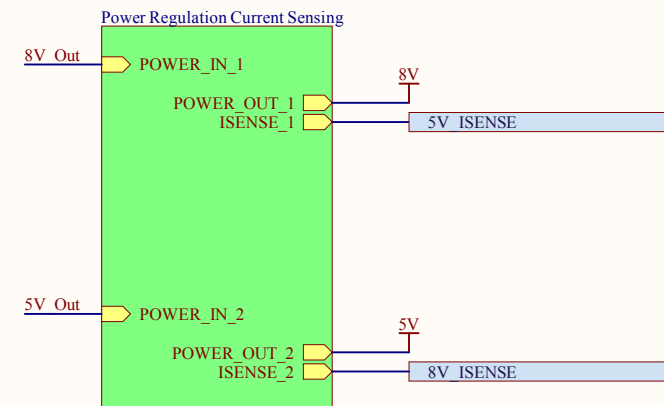
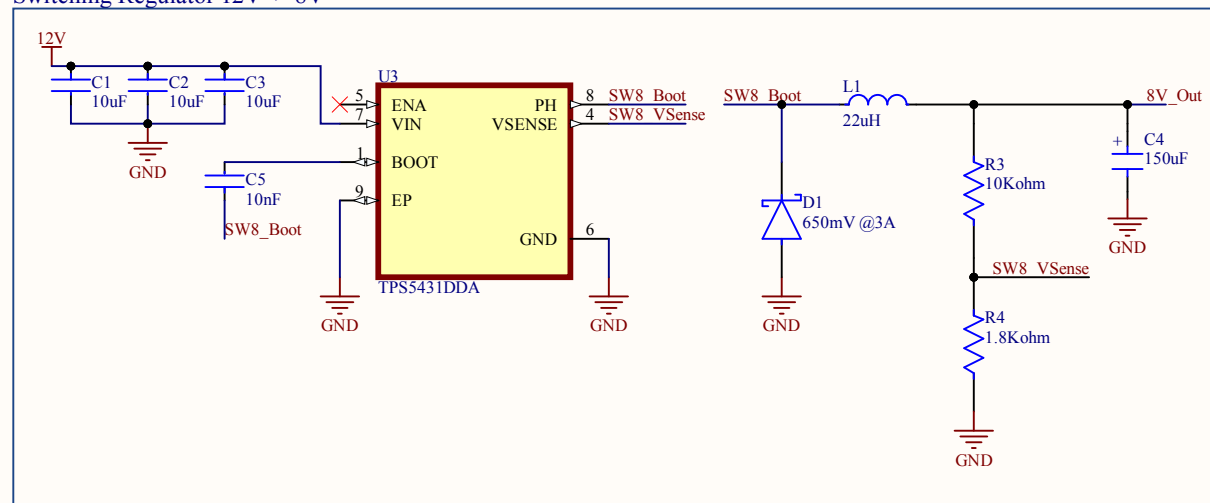
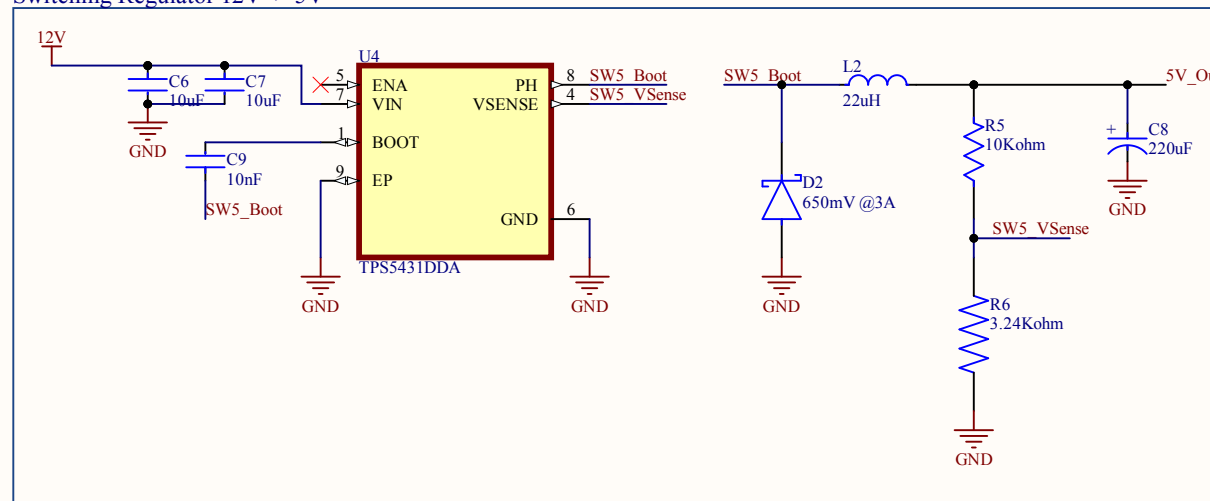




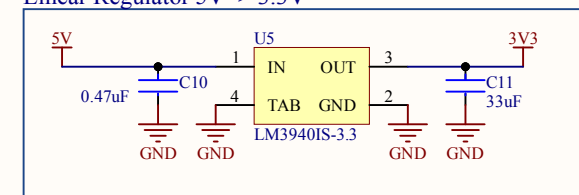
Switching Regulator 12V -> 8V



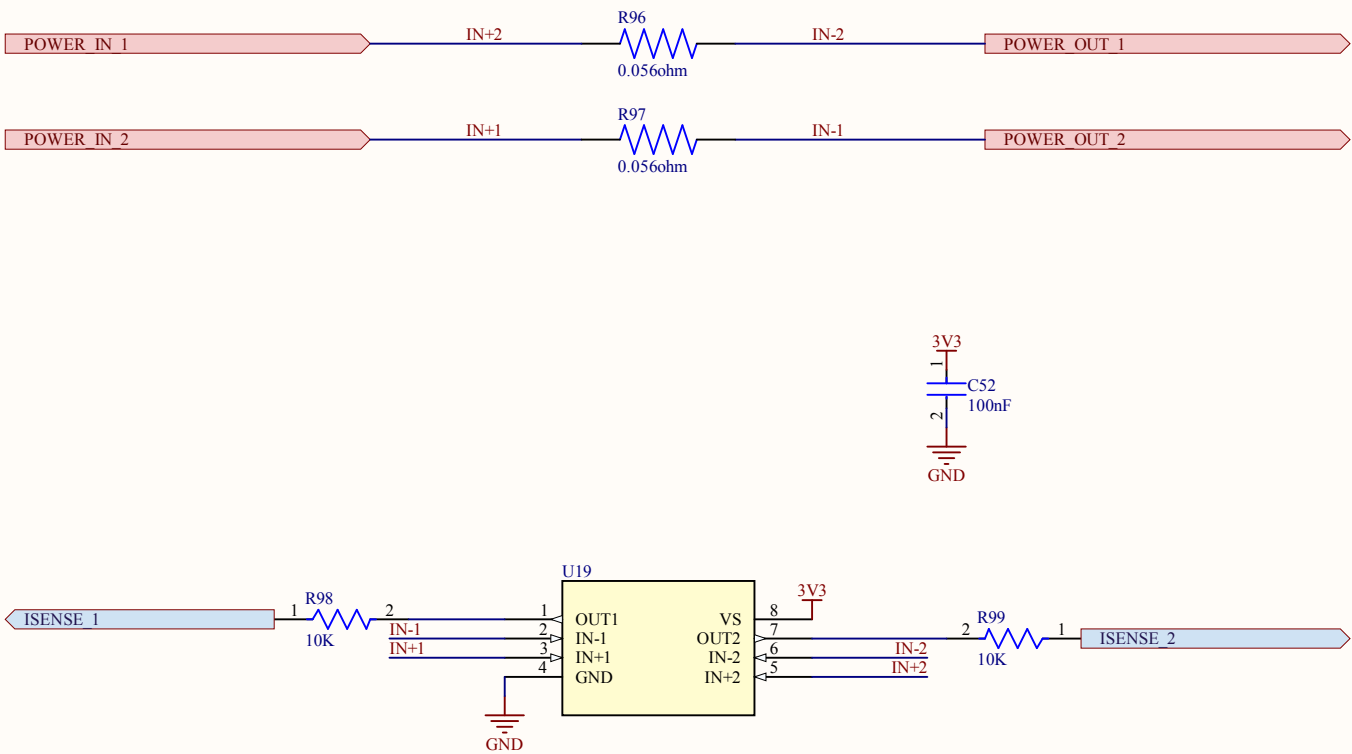
Switching Regulator 12V -> 5V



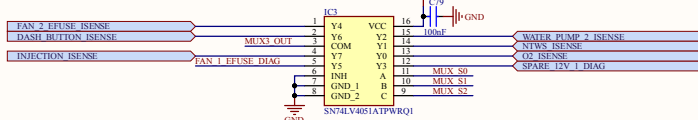
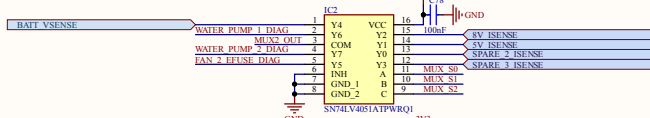
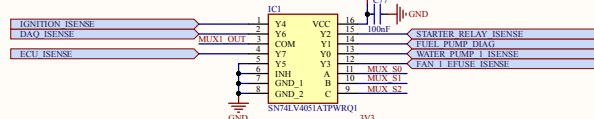
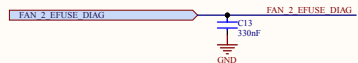
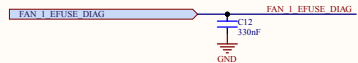
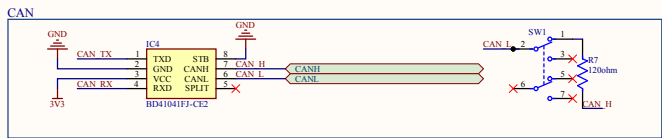
Linear Regulator 5V -> 3.3V



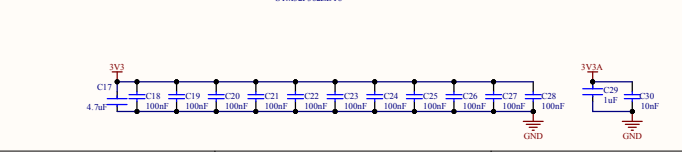
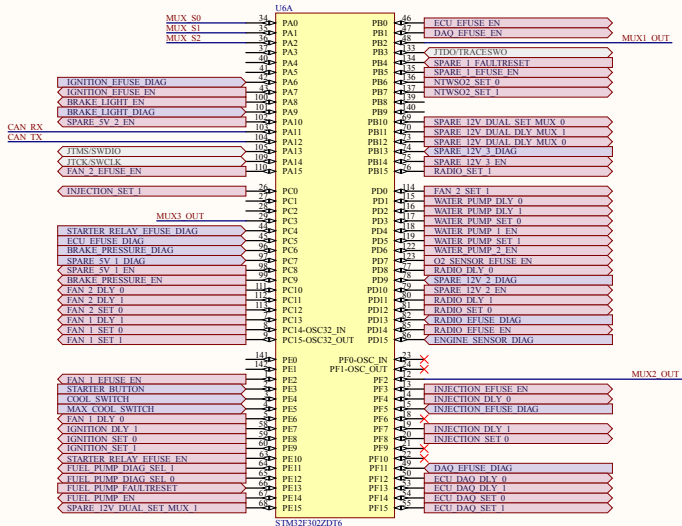
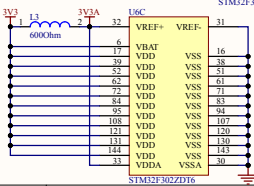
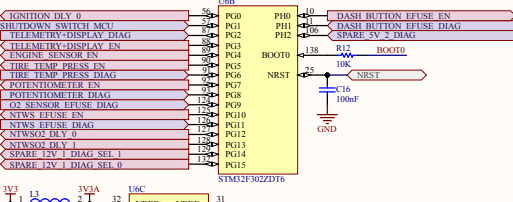
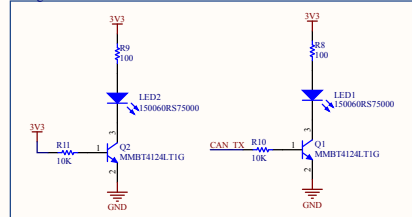
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Power Regulation Schematic		
Size	Number	Revision
A4		
Date:	4-02-2022	Sheet of
File:	C:\Users\...\Power Regulation Module.Sch	Drawn By:



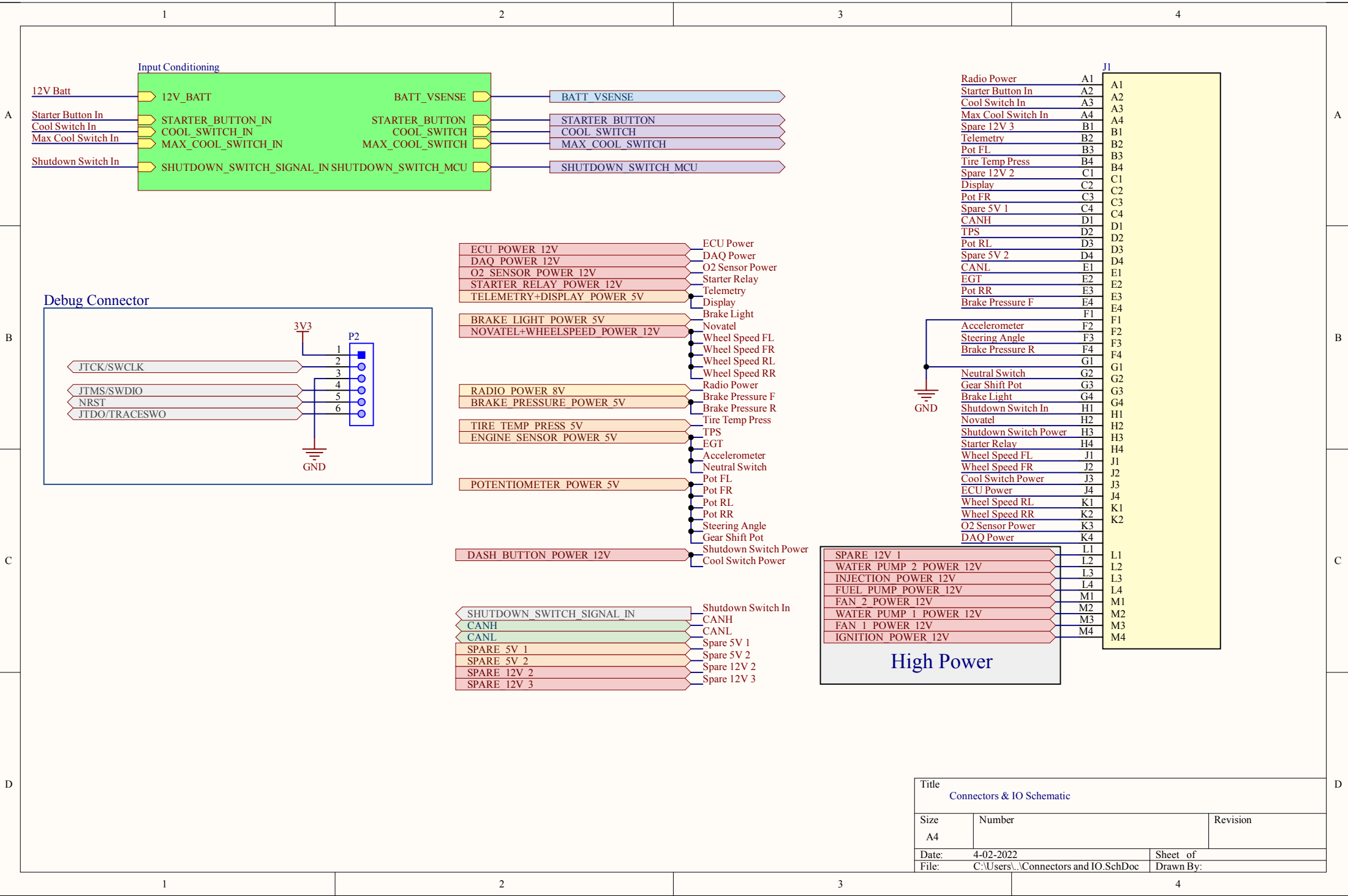
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5V Current Sense		
Size	Number	Revision
A4		
Date:	4-02-2022	Sheet of
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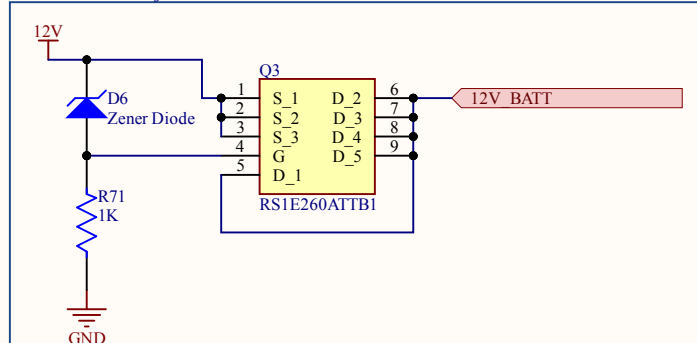
Debug LEDs



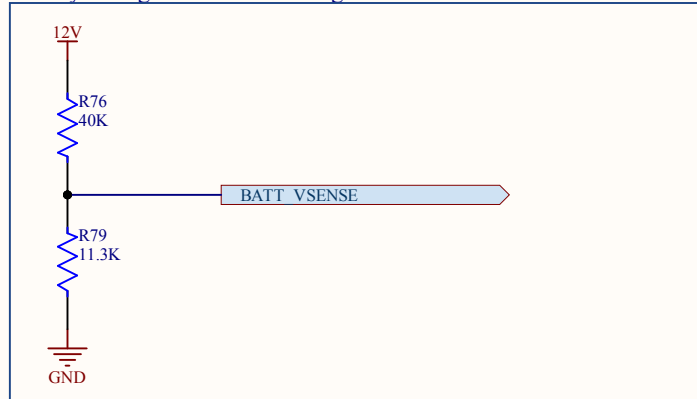
Title			
MCU & CAN Schematic			
Size	Number	Revision	
A2			
Date:	4-02-2022	Sheet of	
File:	C:\Users\MCU SchDoc	Drawn By:	



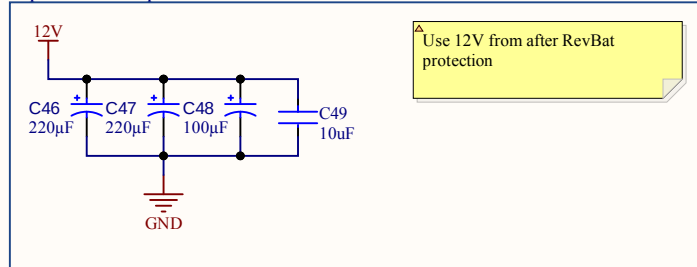
Reverse Battery Protection



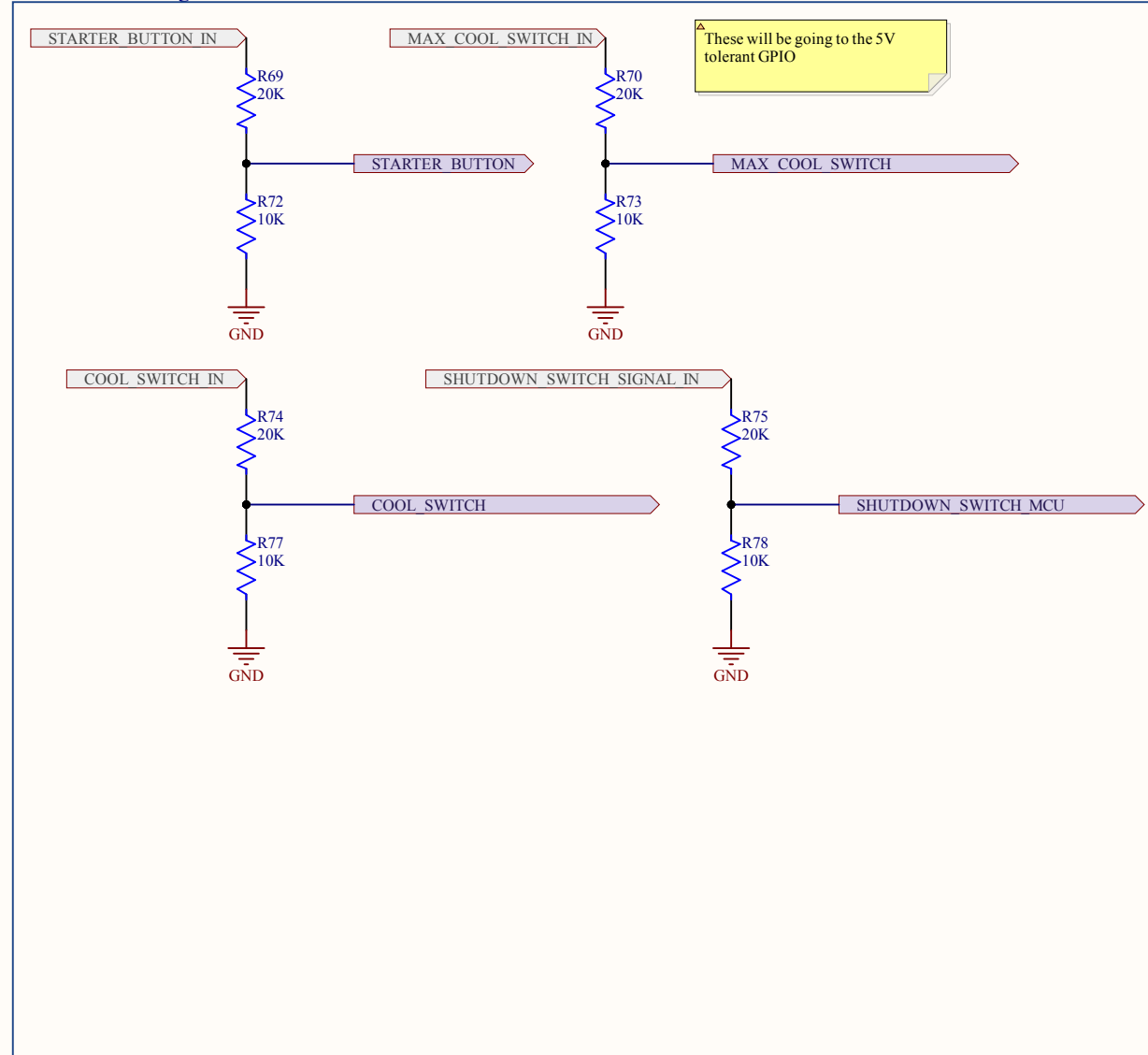
Battery Voltage Divider for Voltage Measurement



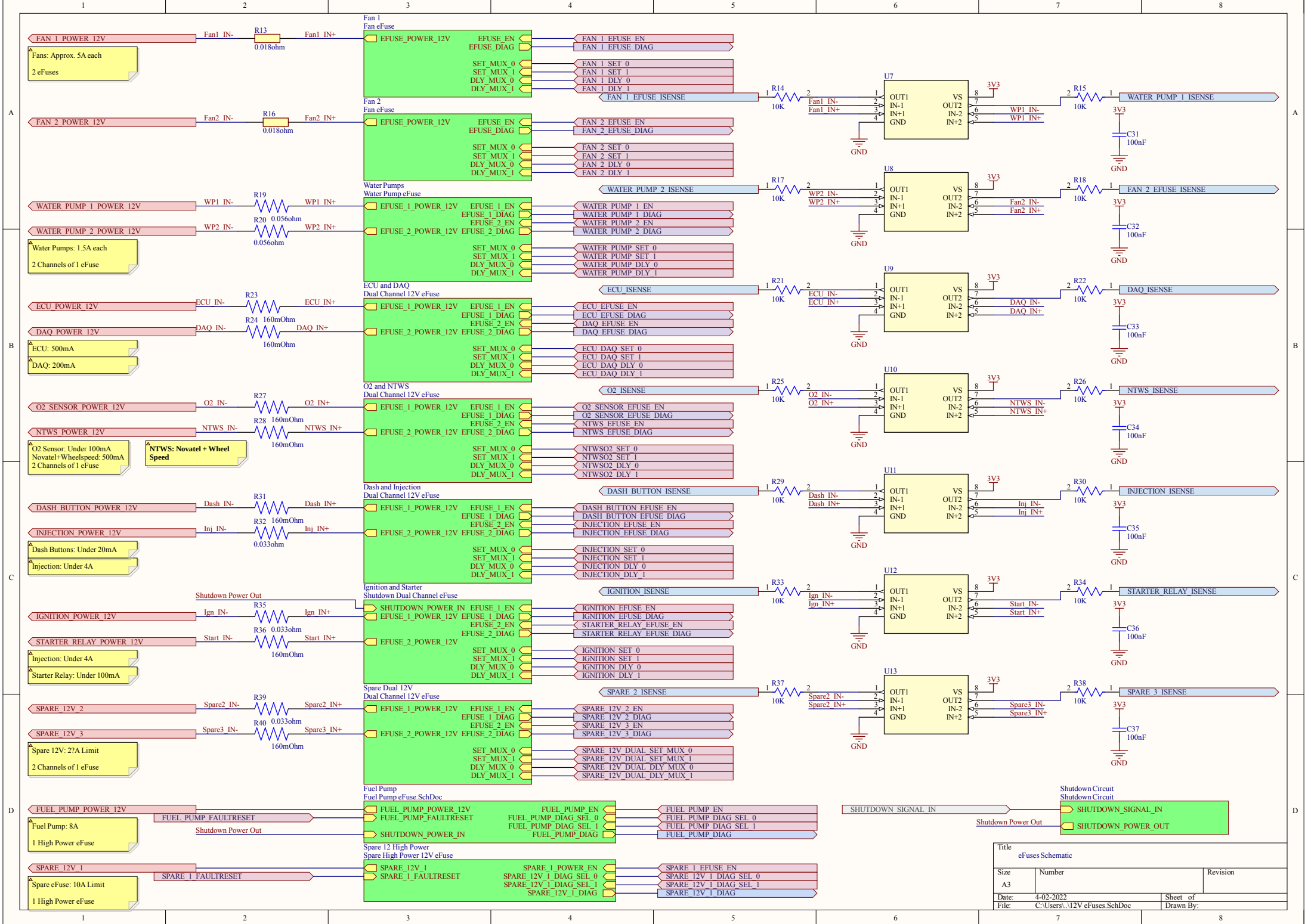
Input Bulk Capacitance



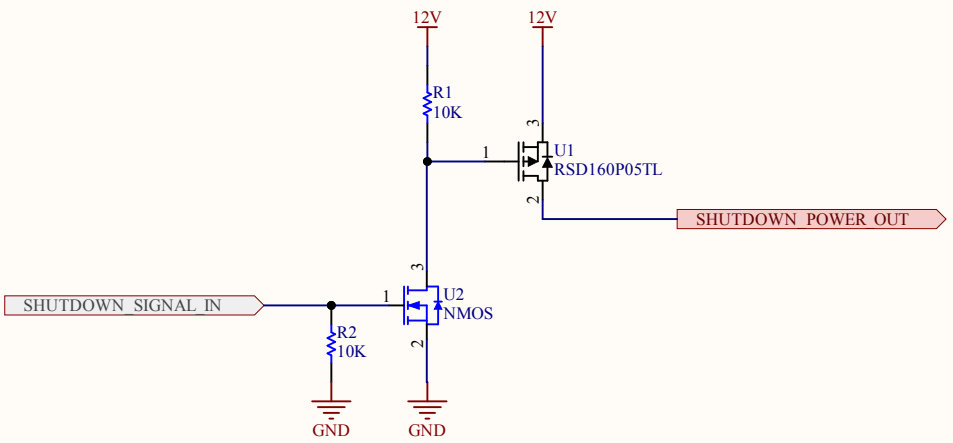
Dash Button Voltage Dividers



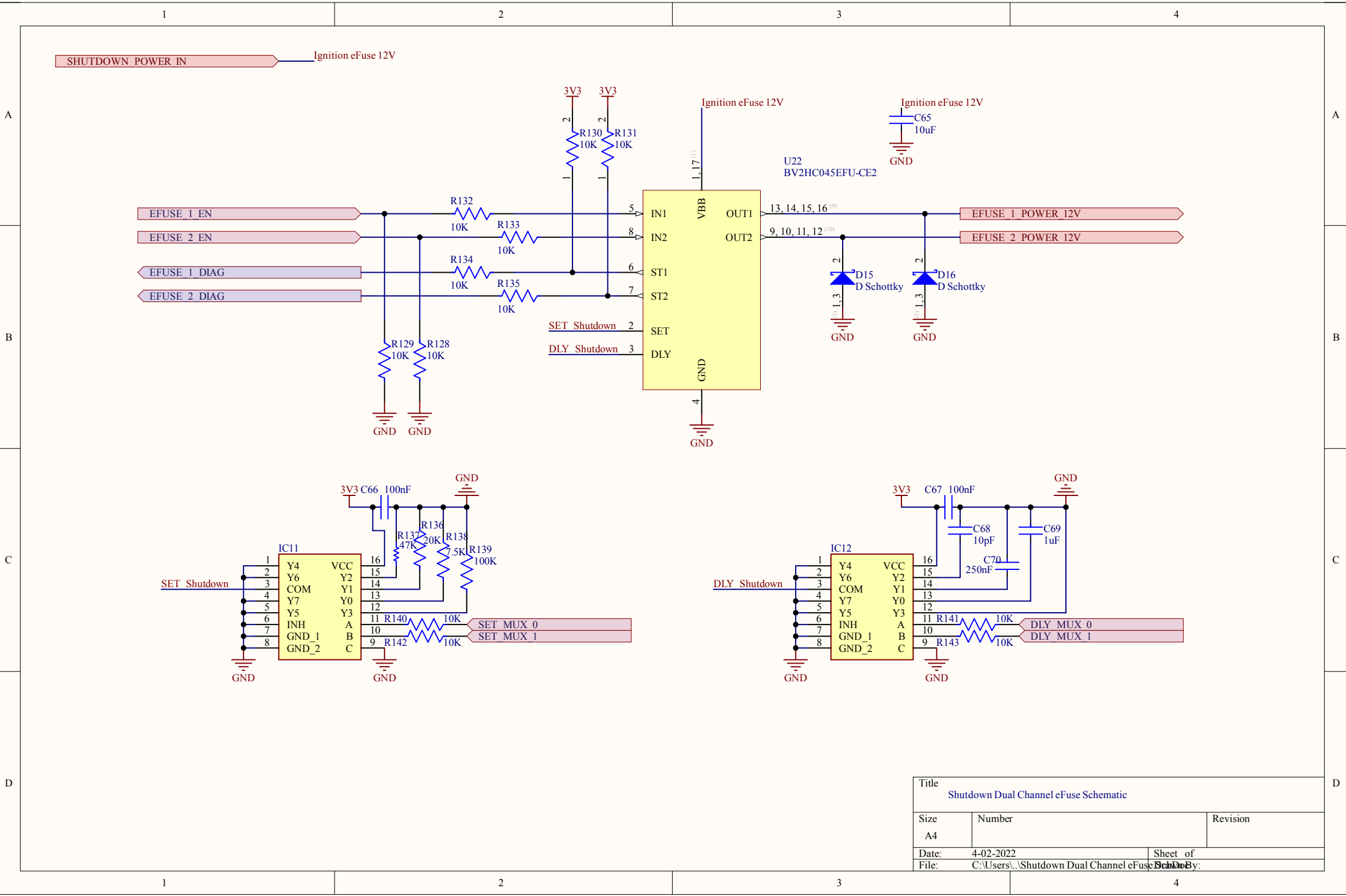
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Size	Number	Revision
A4		
Date:	4-02-2022	Sheet of
File:	C:\Users\...\Input Conditioning.SchDoc	Drawn By:



Title			
eFuses Schematic			
Size	Number	Revision	
A3			
Date:	4-02-2022	Sheet of	
File:	C:\Users\12V eFuses.SchDoc	Drawn By:	



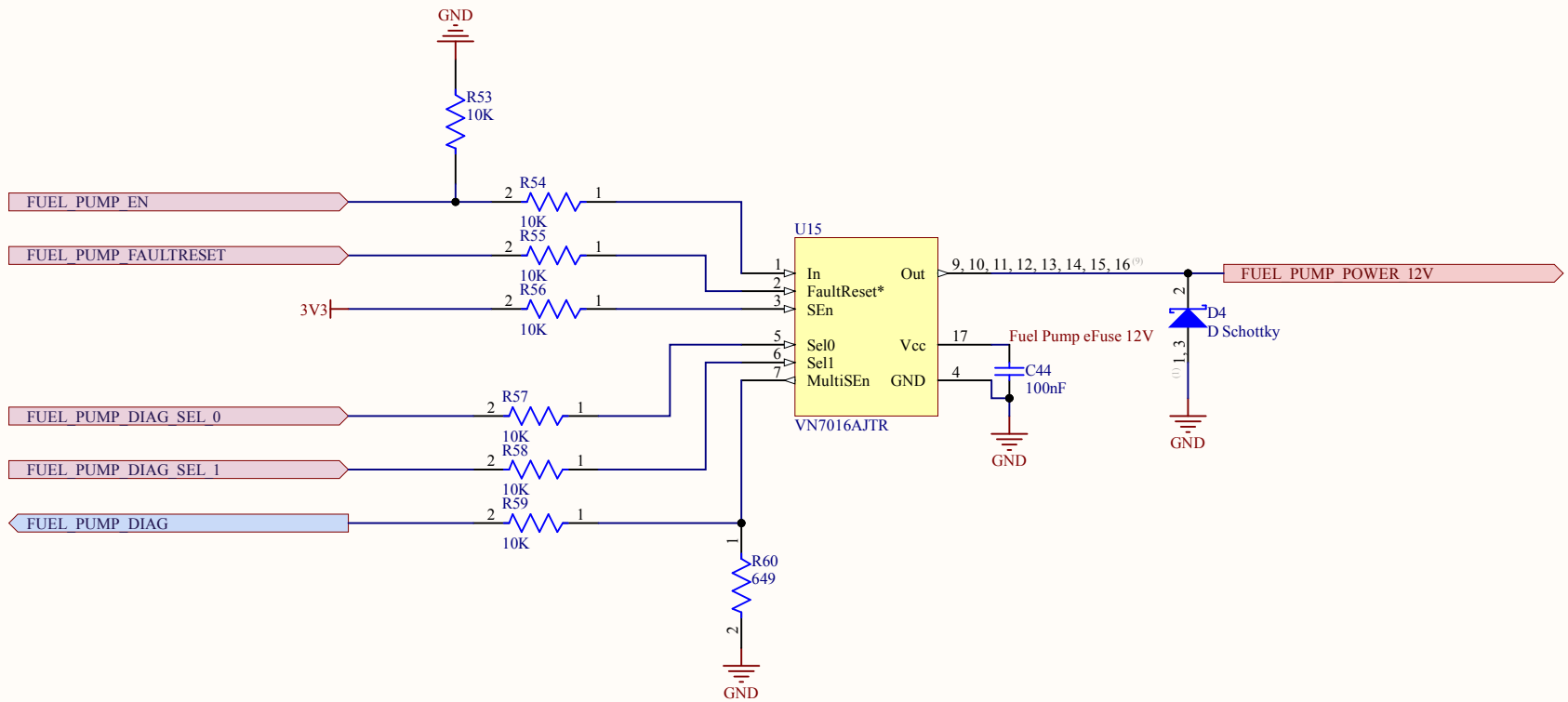
Title Shutdown Circuit		
Size A4	Number	Revision
Date:	4-02-2022	Sheet of
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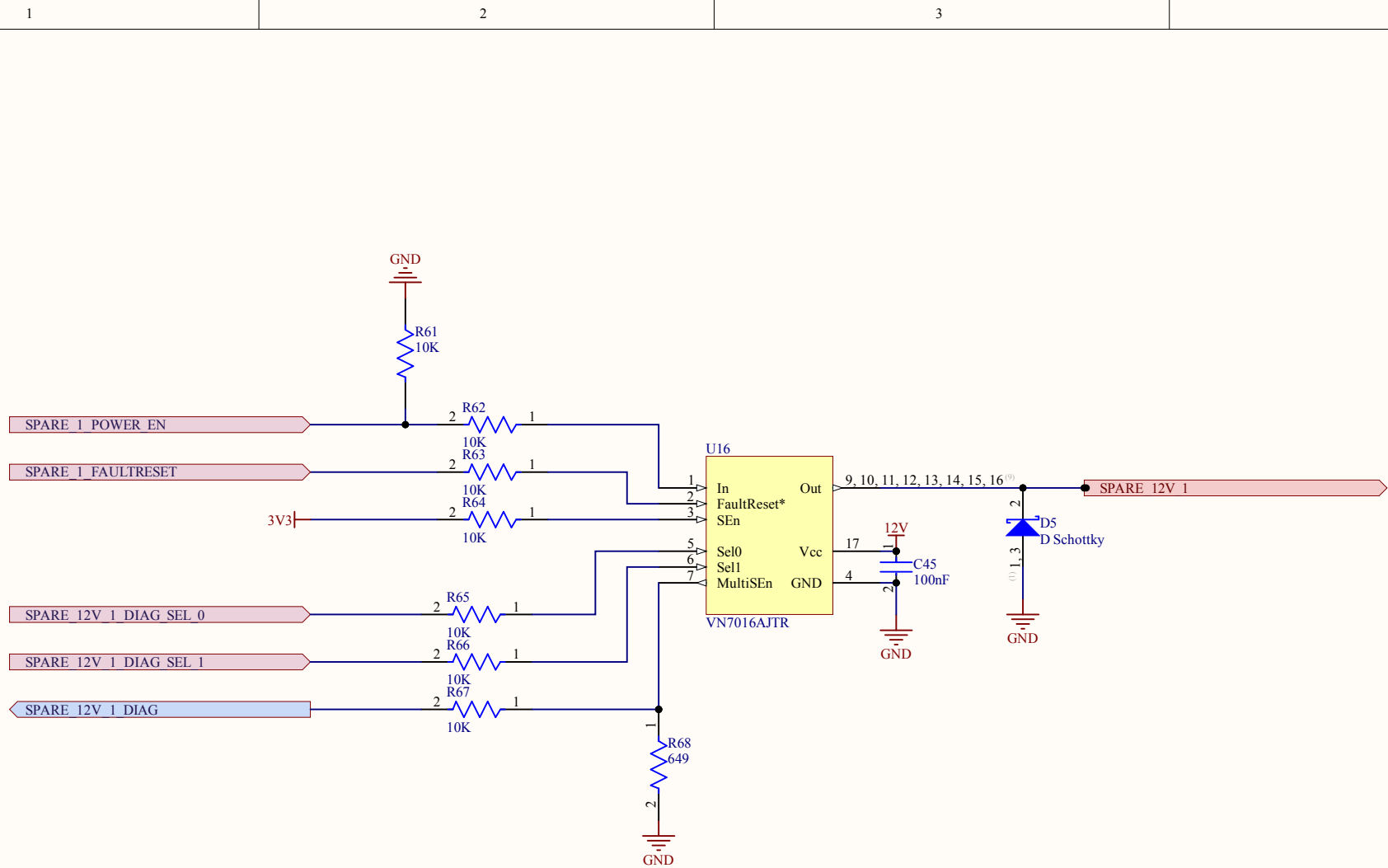
Title		
Shutdown Dual Channel eFuse Schematic		
Size	Number	Revision
A4		
Date:	4-02-2022	Sheet of
File:	C:\Users\...\Shutdown Dual Channel eFuse Schematic	By:

SHUTDOWN POWER IN Fuel Pump eFuse 12V

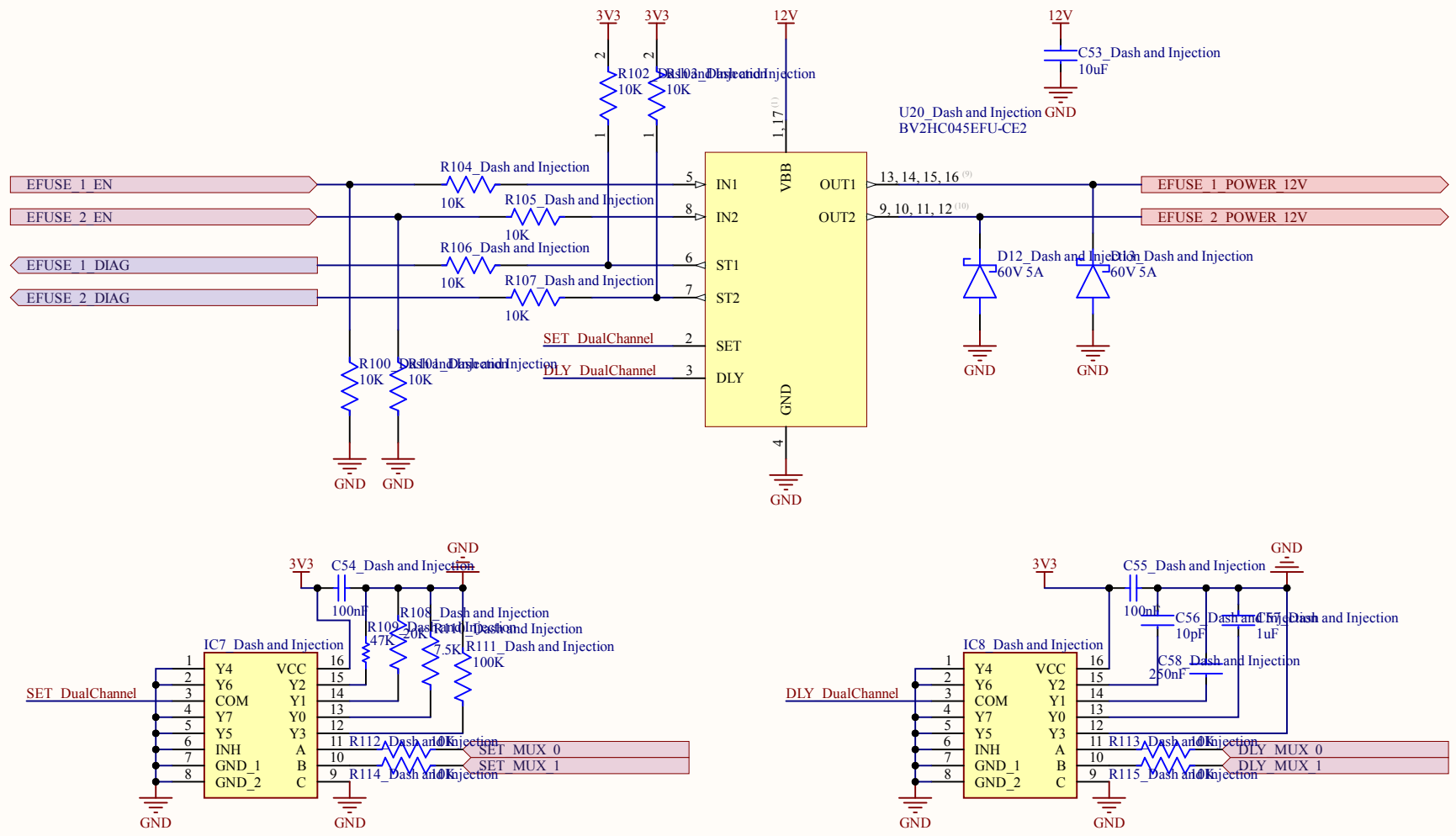
▲ Fuel pump will be disabled when shutdown circuit activated
- Required by rules



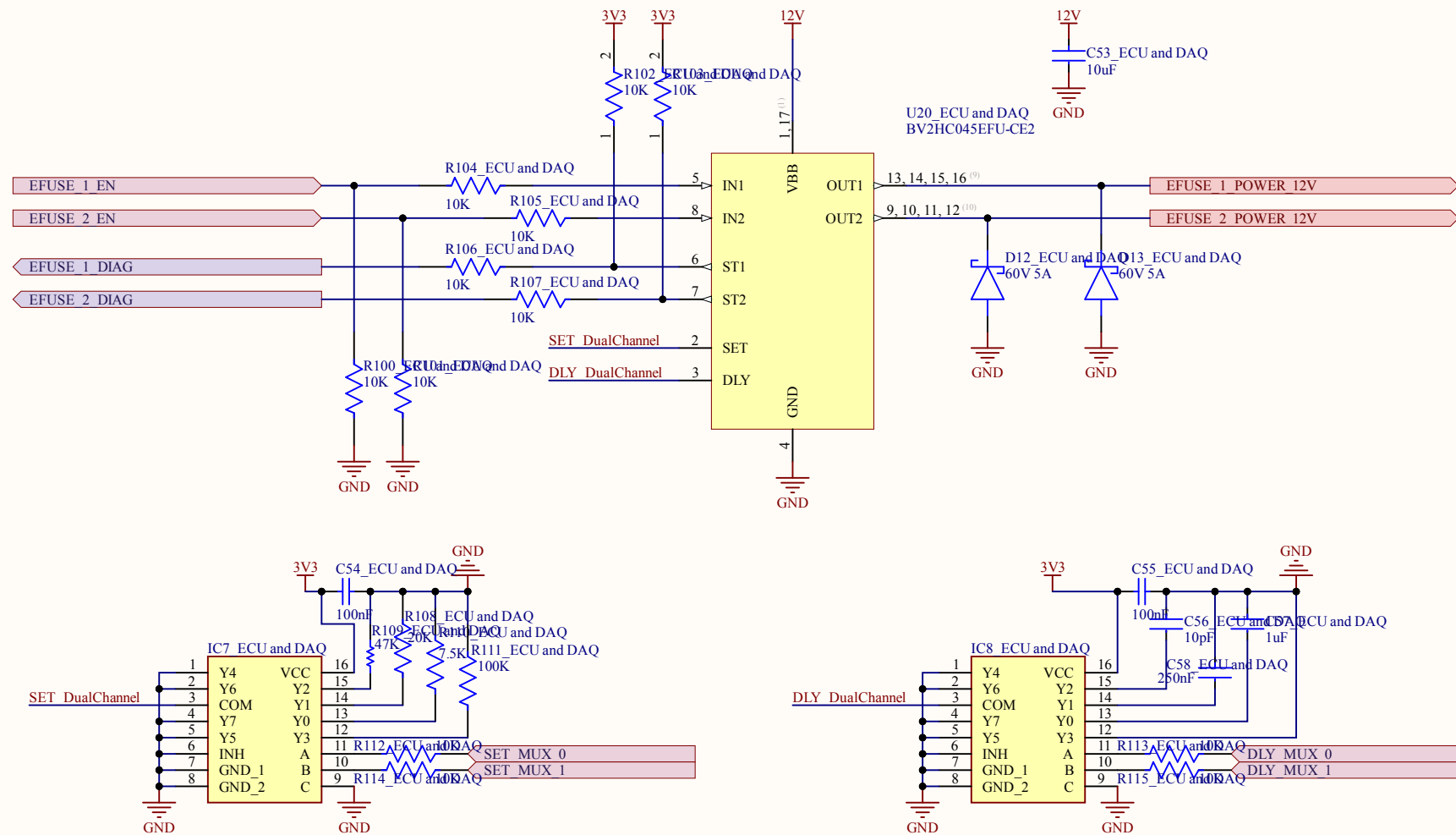
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Size	Number	Revision
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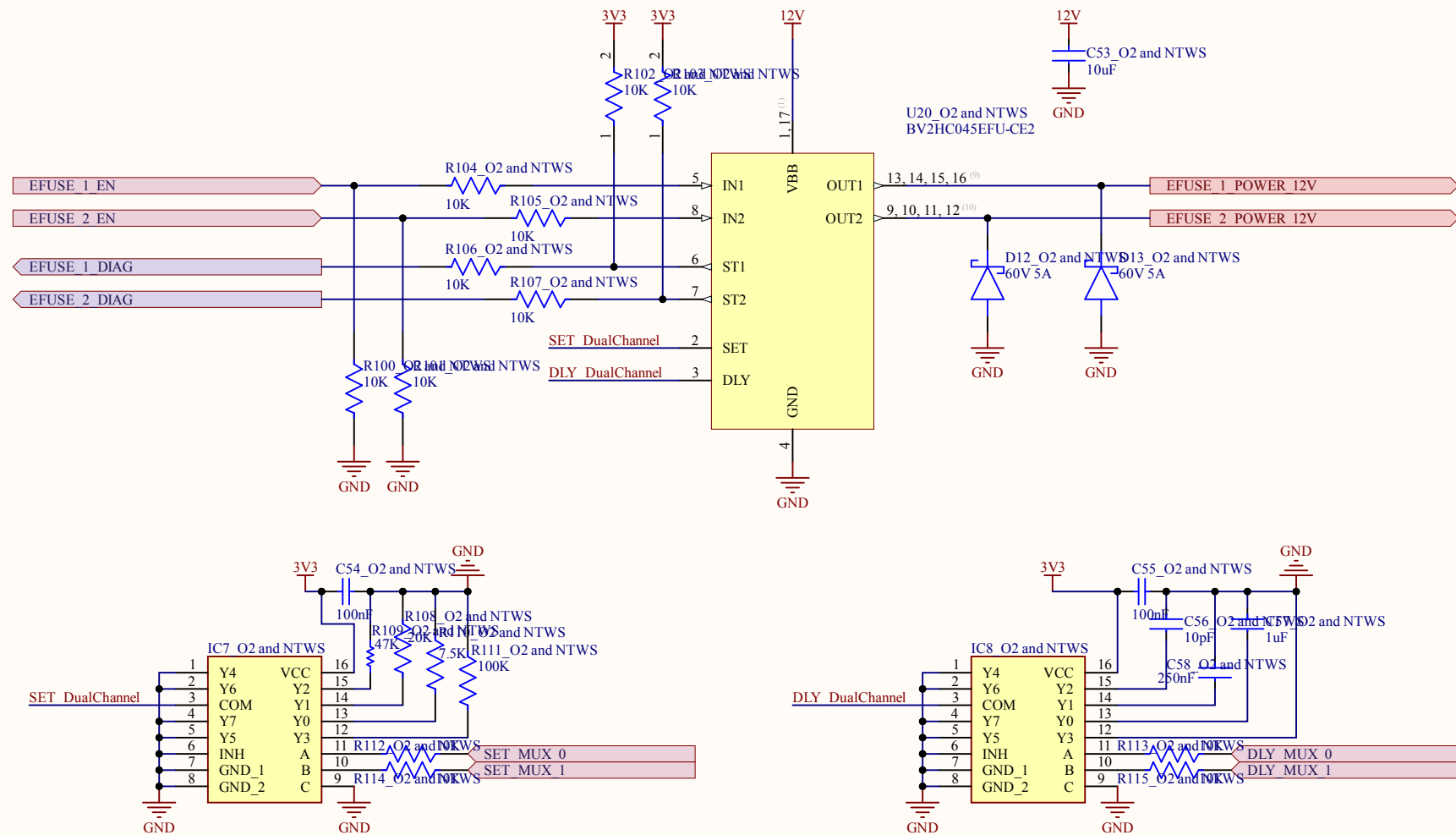
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Spare High Power 12V eFuse Schematic		
Size	Number	Revision
A4		
Date:	4-02-2022	Sheet of
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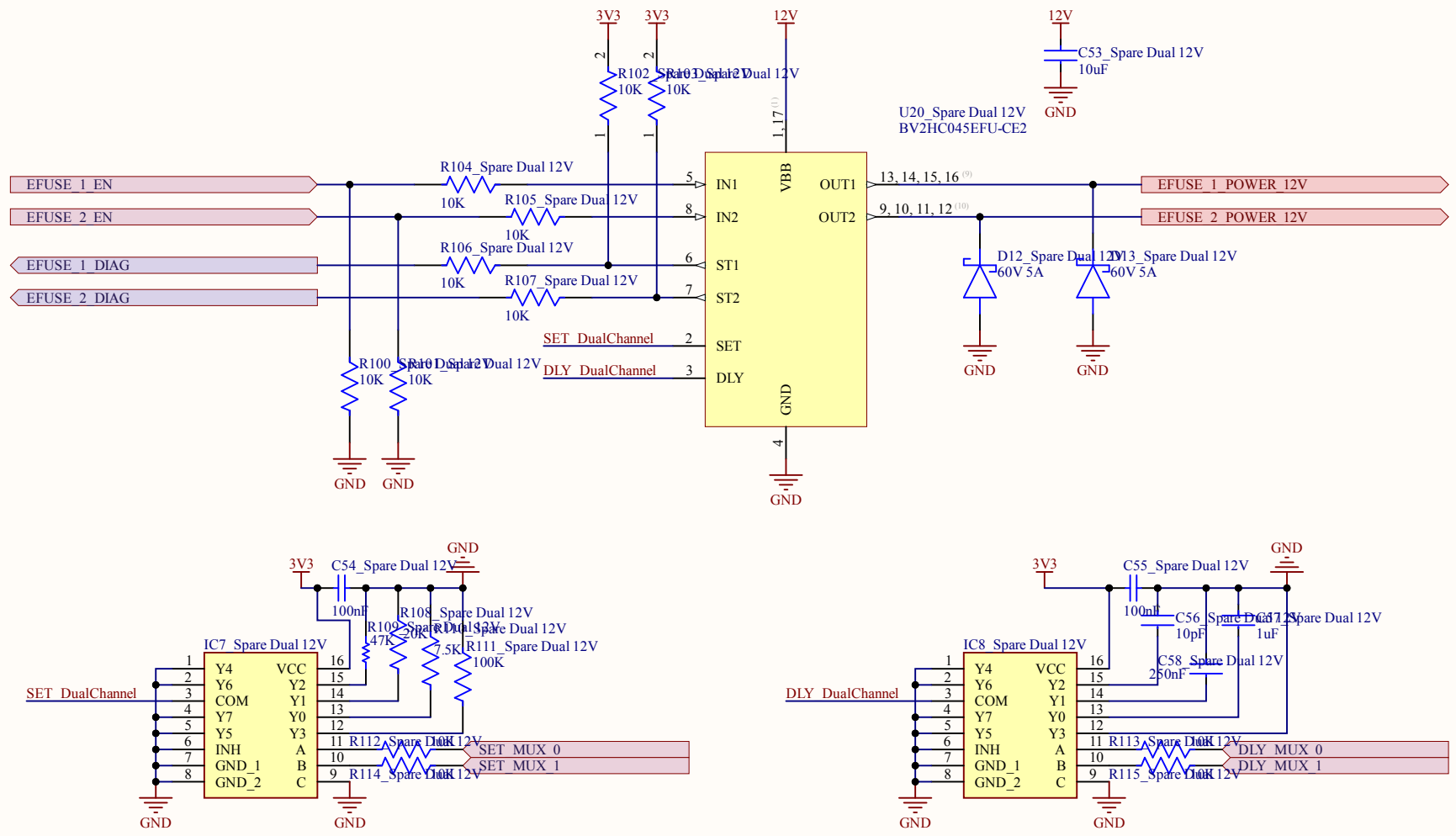
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Size	Number	Revision
A4		
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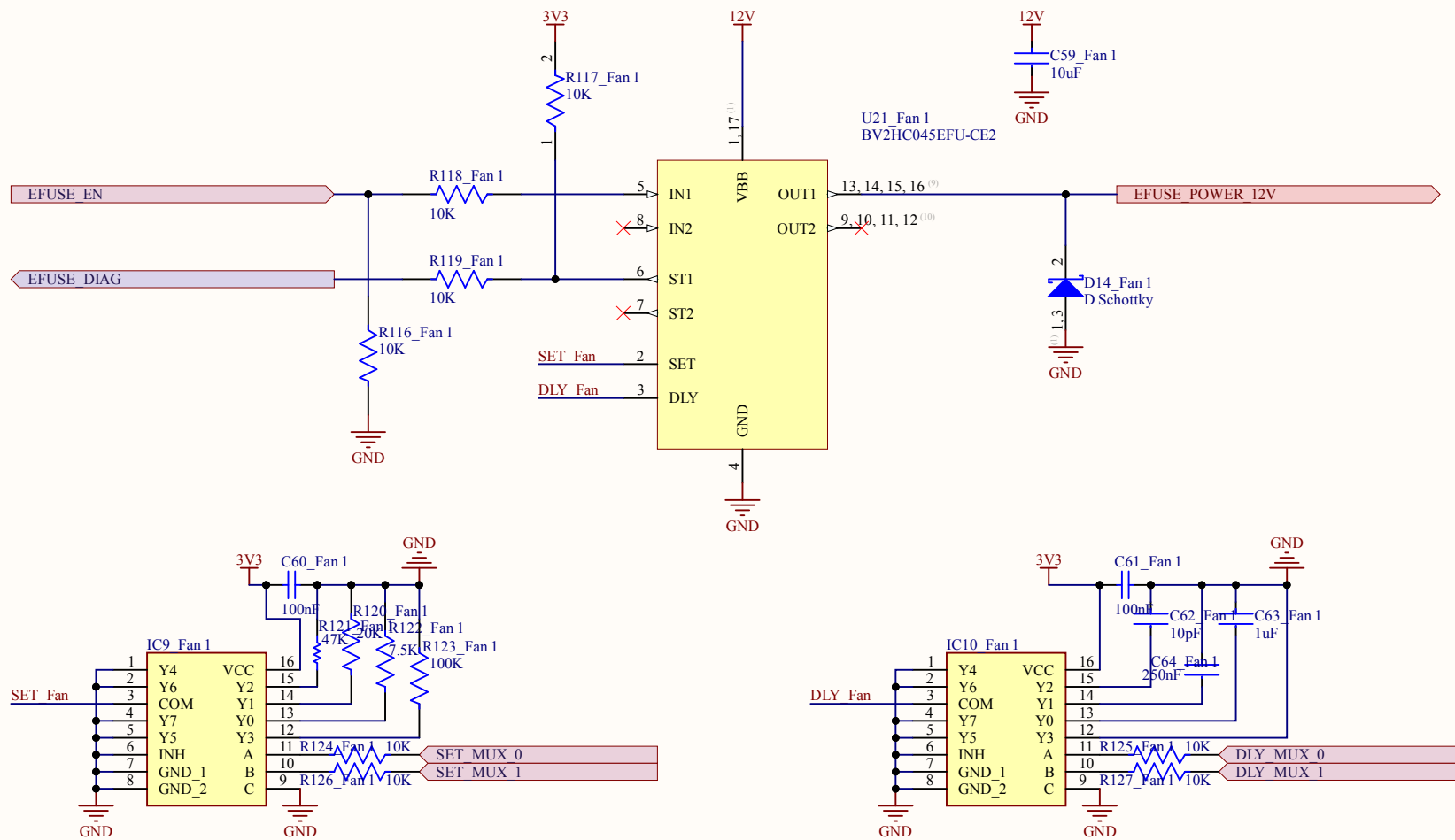
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Size	Number	Revision
A4		
Date:	4-02-2022	Sheet of
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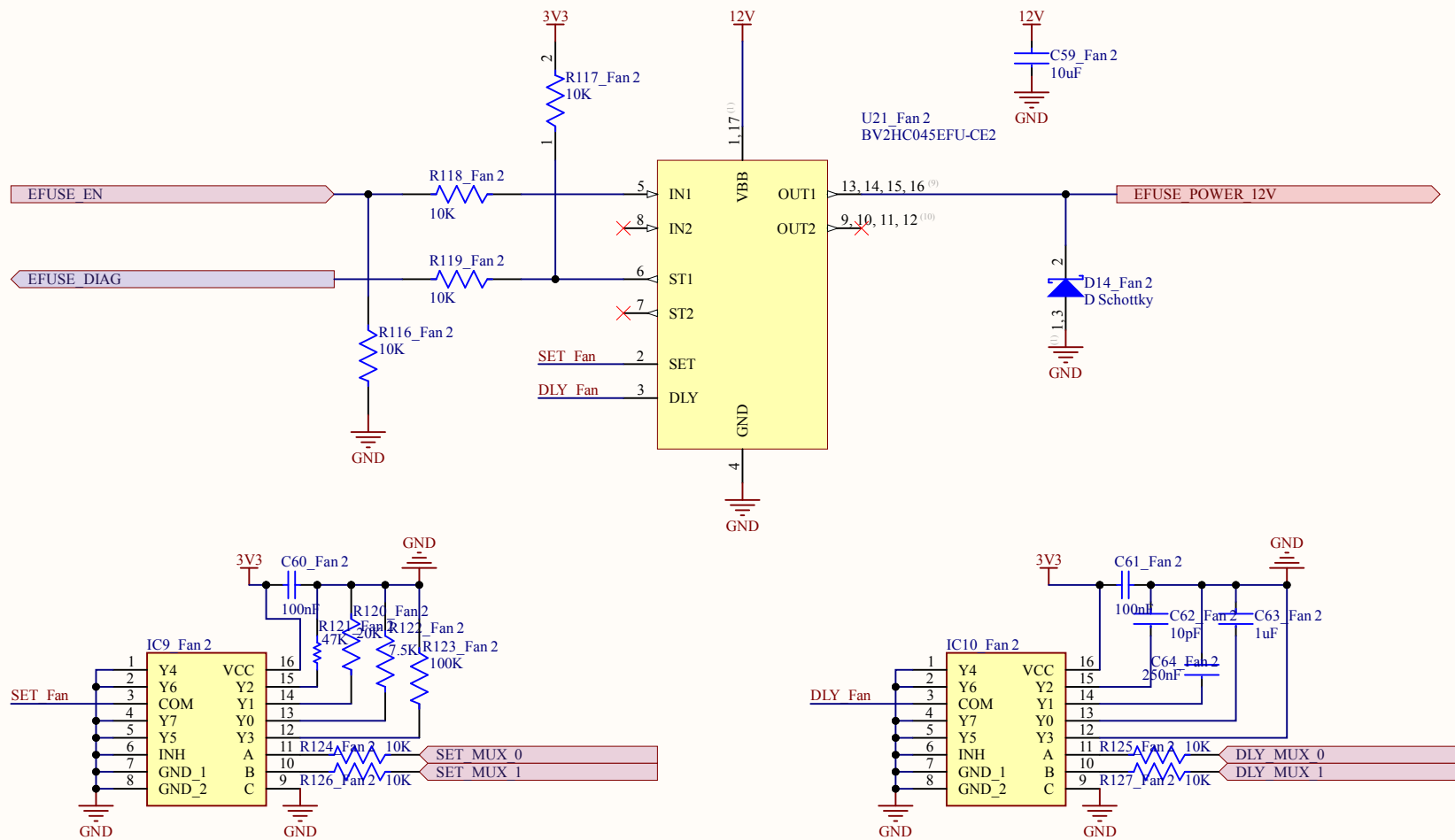
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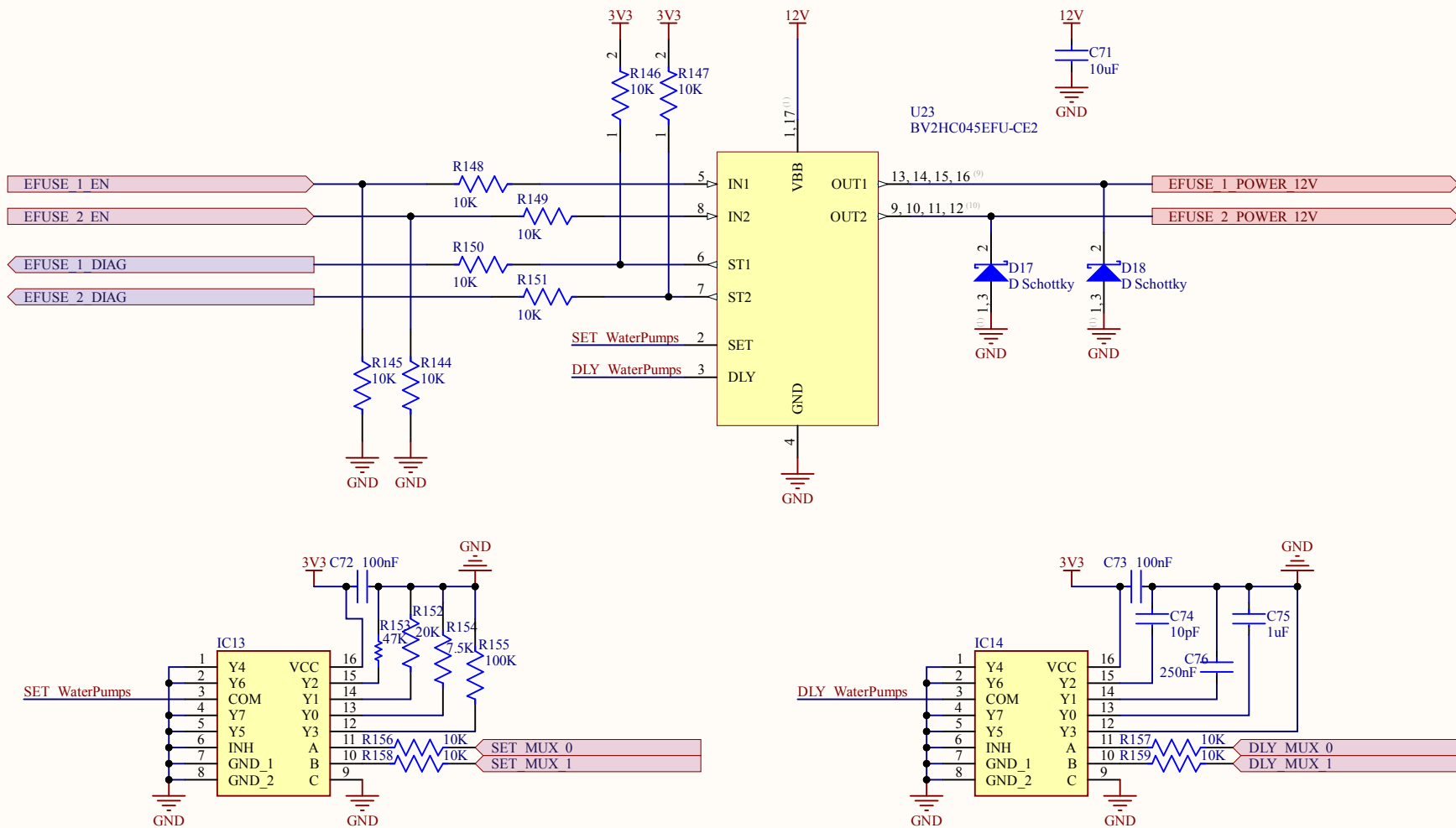
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Size	Number	Revision
A4		
Date:	4-02-2022	Sheet of
File:	C:\Users\...\Dual Channel 12V eFuse.Sch	Drawn By:



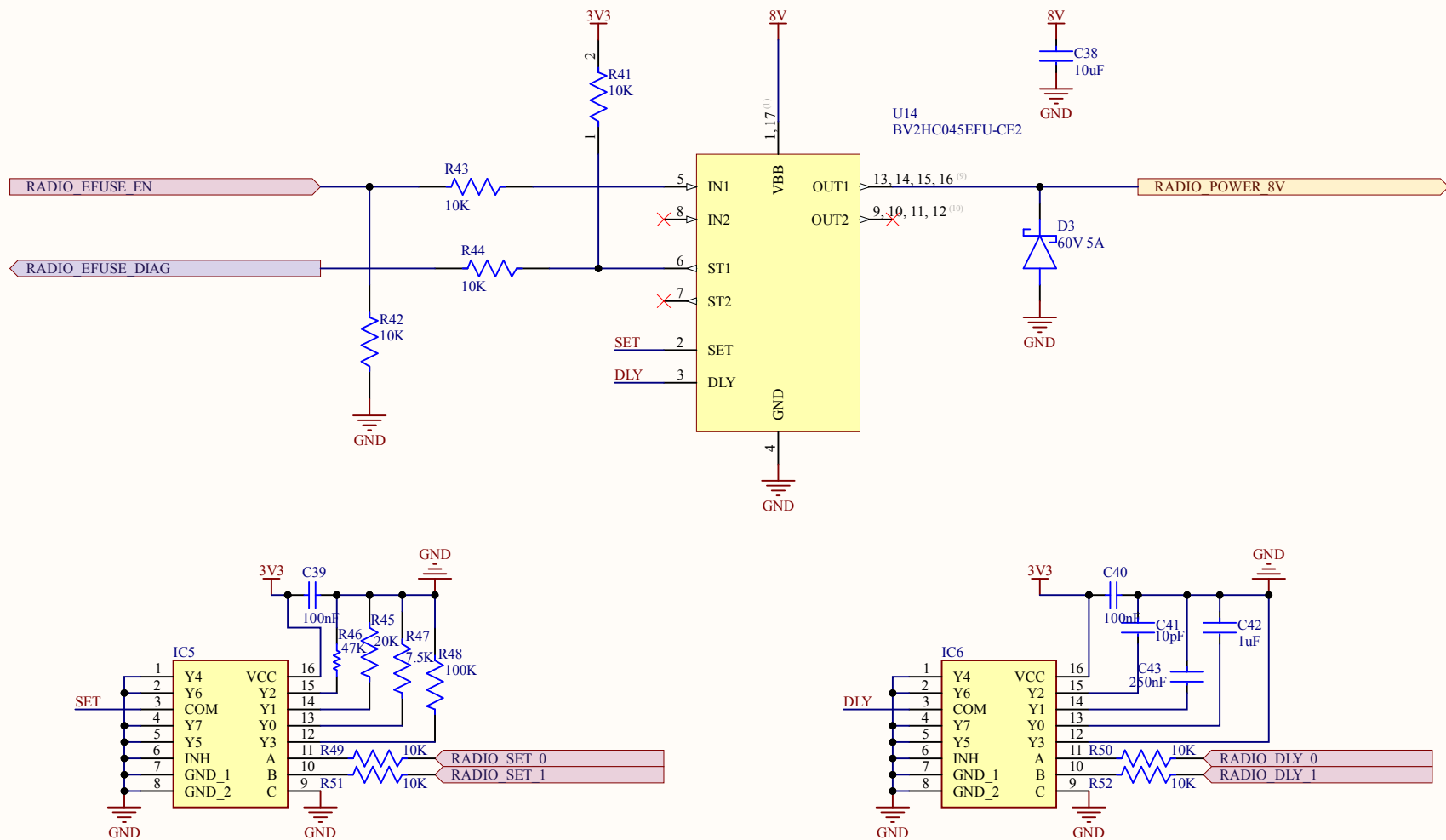
Title		
Fan eFuse Schematic		
Size	Number	Revision
A4		
Date:	4-02-2022	Sheet of
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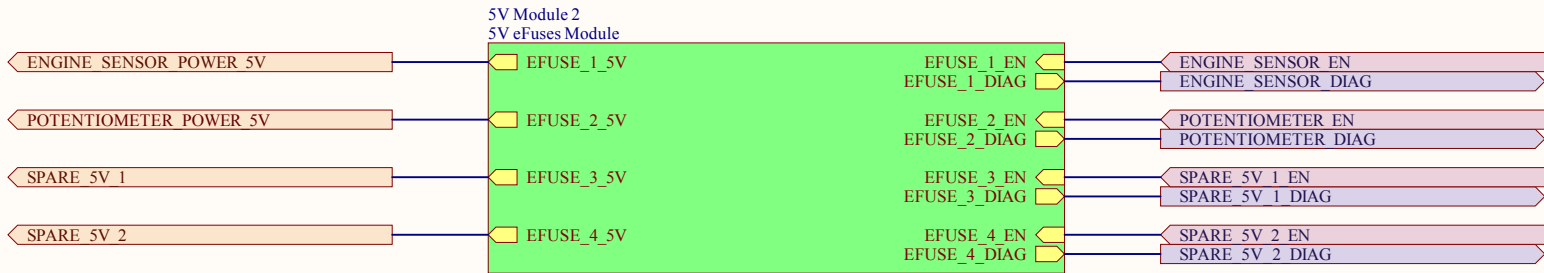
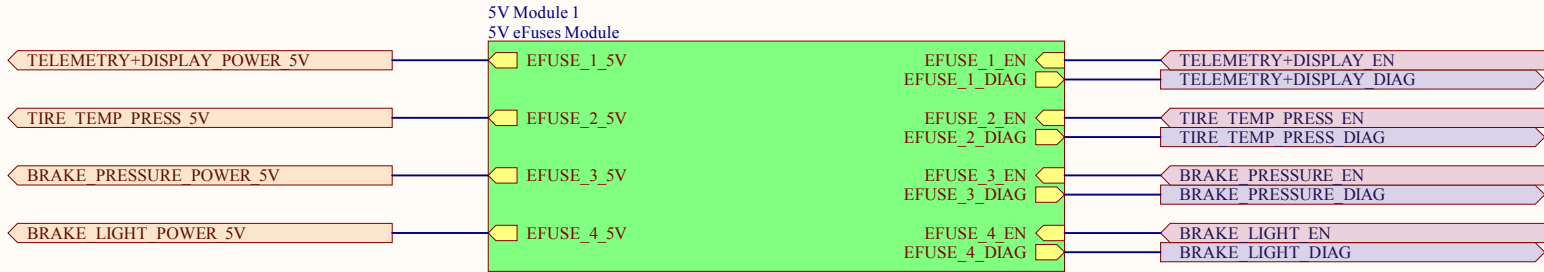
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Size	Number	Revision
A4		
Date:	4-02-2022	Sheet of
File:	C:\Users\...\Fan eFuse.SchDoc	Drawn By:



Title		
Water Pump eFuse Schematic		
Size	Number	Revision
A4		
Date:	4-02-2022	Sheet of
File:	C:\Users\...\Water Pump eFuse.SchDoc	Drawn By:



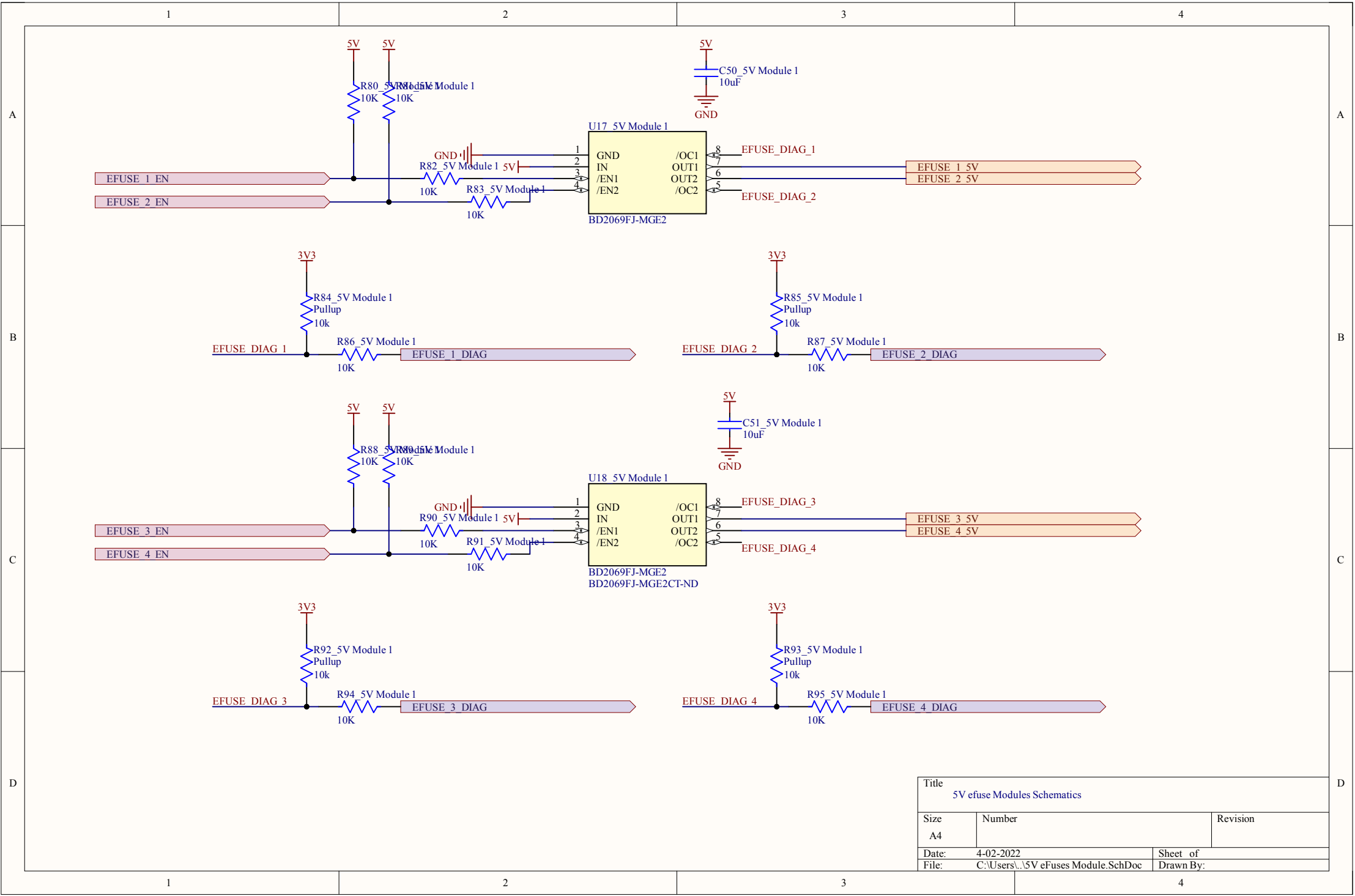
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Size	Number	Revision
A4		
Date:	4-02-2022	Sheet of
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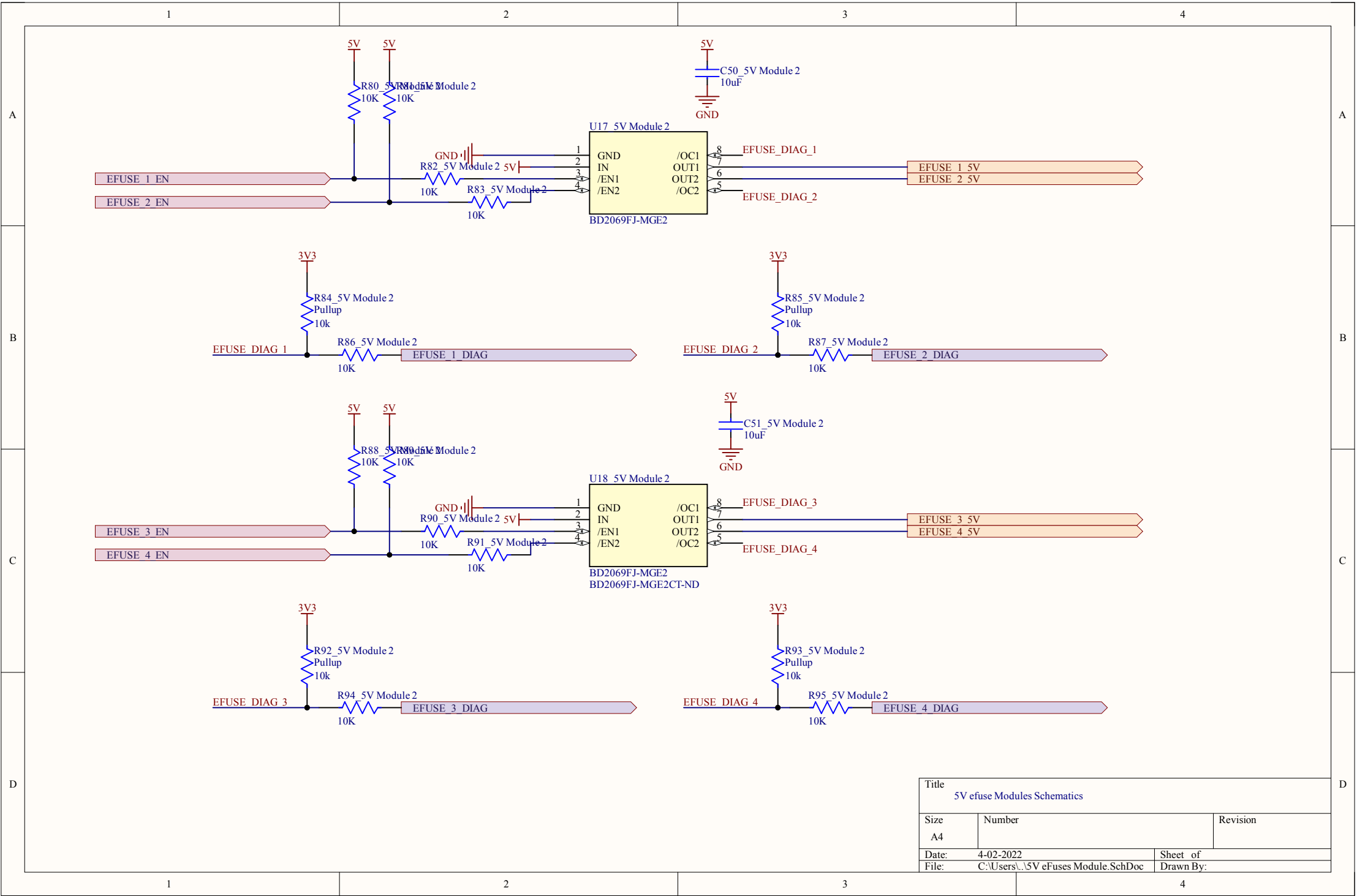
ENGINE_SENSOR_POWER_5V:
- TPS
- EGT
- Accelerometer
- Neutral Switch

POTENTIOMETER_POWER_5V
- Suspension Pots (x4)
- Rotary Pot (Steering Angle)
- Gear Shift Pot

Title		
5V eFuse Schematic		
Size	Number	Revision
A4		
Date:	4-02-2022	Sheet of
File:	C:\Users\...\5V eFuses.SchDoc	Drawn By:



Title		
5V eFuse Modules Schematics		
Size	Number	Revision
A4		
Date:	4-02-2022	Sheet of
File:	C:\Users\...\5V eFuses Module.SchDoc	Drawn By:



Qty #	Name	Description	Designator	Revision ID	Revision Status	Quantity	Manufacturer	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	10uF	CAP CER 10uF 50V X7R 1206	C1, C2, C3, C6, C7, C38, C49, C50, 5V Module 1, C50, 5V Module 2, C53, 5V Module 1, C51, 5V Module 2, C53, Dash and Injection, C53, ECU and DAQ, C53, C53 and NTWS, C53, Spare Dual 12V, C59, Fan 2, C65, C71			10	Samsung	CL18100KBHNNNE		Digi-Key	1276-6767-1-ND	0.248	4.71
	150uF	Cap Tant Polymer 150uF 16VDC Y CASE 20H (7.3 X 4.3 X 2mm) SMD 7343-20 0.05 Ohm 85C T/R	C4			1	Yxcosys AVX	TCJ1517M016R0050	Volume Production	Future Electronics	5088063	0.875	875
	10kF	Cap Ceramic 10kF 50V X7R ±10% Pad SMD 0803 +125°C T/R	C5, C9			2	Samsung	CL10B100KBWJP1C		Digi-Key	1276-CL10B100KBWJPXCCT-ND	0.11	0.22
	220uF	CAP TANT POLY 220uF 6.3V 3411	C8			3	Panasonic	6TPE220MA2B	Volume Production	Digi-Key	P36245CT-ND	1.77	1.77
	0.47uF	CAP CER SMD 0603 .47uF 5% X7R 16	C39			1	AVX	C0603C474RACALTO	Volume Production	Avnet	764H4859	0.10073	0.10073
	33uF	Multilayer Ceramic Capacitor 33uF ±20% 10V X5R SMD 0805	C11			1	TDK	C332X33A18MAG2AC	Volume Production	Digi-Key	445-6236-1-ND	0.94	0.94
	330uF	Cap Ceramic 330uF 50V X7R ±20% Pad SMD 0803 +125°C T/R	C12, C13, C34, C35			1	Samsung	CL18330KBHNNNE		Mouser	187-CL18330KBHNNNN	0.03	0.3
	100uF, 10kF	C18, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C30, C31, C32, C33, C34, C35, C36, C37, C38, C40, C44, C44, Dash and Injection, C54, ECU and DAQ, C54, C2 and NTWS, C54, Spare Dual 12V, C55, Dash and Injection, C55, ECU and DAQ, C55, C2 and NTWS, C55, Spare Dual 12V, C59, Fan 1, C60, Fan 2, C65, Fan 1, C61, Fan 2				20	Yxcosys AVX	0608C10K4Z2A	Volume Production	Digi-Key	478-1590-1-ND	0.0306	1.98
	GRM21BR71C479K5XK	Chip Multilayer Ceramic Capacitors for General Purpose, 0805, 4.7uF, X7R, 15%, 10%, 16V	C17			1	Murata	GRM21BR71C479K5XK		Digi-Key	482-GRM21BR71C479K5XKCT-ND	0.19	0.19
	1uF	Cap Ceramic 1uF 25V X7R ±10% Pad SMD 0803 +125°C T/R	C26, C42, C57, Dash and Injection, C57, ECU and DAQ, C57, C57 and NTWS, C57, Spare Dual 12V, C63, Fan 1, C63, Fan 2, C69, C75			10	Samsung	CL10B100KA9FNC	Volume Production	Mouser	187-CL10B100KA9FNC	0.027	0.27
	06035A100TA4A, 10kF	10uF ±5% 50V Ceramic Capacitor, [N/A]	C41, C56, Dash and Injection, C56, ECU and DAQ, C56, C56 and NTWS, C56, Spare Dual 12V, C62, Fan 1, C62, Fan 2, C68, C74			8	AVX	C0603C100X3HCAUTO		Digi-Key	369-C0603C100X3HCAUTOCT-ND	0.082	0.65
	250uF	0.21 Series 0805 250uF 25V ±10% Tolerance X7R Multilayer Ceramic Chip Capacitor	C43, C58, Dash and Injection, C58, ECU and DAQ, C58, C58 and NTWS, C58, Spare Dual 12V, C59, Fan 1, C59, Fan 2, C70, C76			6	Samsung	CL2B224KAFNNGN	Volume Production	Mouser	187-CL2B224KAFNNGN	0.034	0.34
	100uF	020903 0.10uF X7R 300pm/C 10.00% 25V	C45, C52, C66, C67, C72, C73, C77, C78, C79			1	AVX	C0603C10K39AC7081		Newark	87AH4904	0.015	0.15
	ECA-1HM21	Aluminum Electrolytic Capacitor, 220uF, +/- 20%, 50V, -40 to 85 deg., 2-Pin THD, RoHS, Bulk	C46, C47			2	Panasonic	ECA-1HM21	Volume Production	RSComponents	2286919	0.734	3.67
	ECA3HM102	Aluminum Electrolytic Capacitor, 100uF, +/- 20%, 50V, -40 to 85 deg., 2-Pin THD, RoHS, Bulk	C48			1	Panasonic	ECA-3HM102	Volume Production	RSComponents	2286923	0.334	1.67
	650mV @3A	DIODE SCHOTTKY 60V 3A SOD123+6, 650mV @3A	D1, D2			2	ON Semiconductor / Fairchild	FSV30FP	Volume Production				
	60V 5A	Schottky Diode	D15, D15, Dash and Injection, D15, ECU and DAQ, D15, C2 and NTWS, D15, Spare Dual 12V, D13, Dash and Injection, D13, ECU and DAQ, D13, C2 and NTWS, D13, Spare Dual 12V			8	ON Semiconductor / Fairchild						
	D Schottky	Schottky Diode	D4, D5, D14, Fan 1, D14, Fan 2, D15, D16, D17, D18			1	ON Semiconductor	52MM5218V71G	Volume Production	Digi-Key	52MM5218V71G50CT-ND	0.5	0.5
	Zener Diode	500 mW Standard Tolerance Zener Diode Voltage Regulators 18 V (Automotive)	D6			1	Texas Instruments	SN74V4051ATPWRQ1	Volume Production	Digi-Key	296-1590-1-ND	0.71	14.91
	SN74V4051ATPWRQ1	Automotive 5-V, B, 1, 1-channel general-purpose analog multiplexer 12-TSPQ -40 to 105	IC1, IC2, IC3, IC5, IC6, IC7, Dash and Injection, IC7, ECU and DAQ, IC7, C2 and NTWS, IC7, Spare Dual 12V, IC8, Dash and Injection, IC8, ECU and DAQ, IC8, C2 and NTWS, IC8, Spare Dual 12V, IC9, Fan 1, IC9, Fan 2, IC10, Fan 1, IC10, Fan 2, IC11, IC12, IC13, IC14			1	Texas Instruments	SN74V4051ATPWRQ1	Volume Production	Digi-Key	296-1590-1-ND	0.71	14.91
	BD4104UF-C2	CAN Transceiver 1Mbps Normal/Standby 4.75V to 5.25V 8-Pin SOP-J Emboss T/R	IC4			1	Robm	BD4104UF-C2E2		Digi-Key	846-BD4104UF-C2E2CT-ND	1.68	1.68
	36038002	2.50mm, 3.50mm DMC Header, Right-Angle, 48 Position, 1 Port, Black, Peg, Lead-Free	J1			1	Molex	36038-0002	Volume Production	RSComponents	7278034	11.81	11.81
	22uH	Commercial Inductors Low DCR Series 22uH 4 3A 94 20mD 20m	L1, L2			2	Vishay Dale	H4P3240-70220M11	Volume Production	Newark	1778073	0.62	1.62
	4600Ohm	SMD EMI Suppression Ferrite Bead WE-CBF HF, Z = 600 Ohm	L3			1	Würth Electronics	728K62160	Volume Production	Digi-Key	728-2385-1-ND	0.23	0.23
	150000R010000	SMD mono-resistor Chip LED, 1W, 40mA, 250mW	LED1, LED2			1	Würth Electronics	150000R010000	Volume Production	Digi-Key	728-4070-1-ND	0.15	0.15
	70uH 102-QT-F-5	0.025" SQ Post Header, Through-Hole, Vertical, -55 to 125 deg., 2.54mm Pitch, 3-Pin, Male, RoHS	F1			1	Antec	70uH-102-QT-F-5	Volume Production	Arrow Electronics	70uH-102-QT-F-5	0.15	0.15
	61300H1121	THT Vertical Pin Header 160-Pin, 0.1in Pitch, 2.54mm, Single Row, 6pins	F2			1	Würth Electronics	61300H1121	Volume Production	Digi-Key	728-5319-1-ND	0.36	0.36
	MMBT4242A,TIG	General Purpose Transistor, NPN Silicon, 3-Pin SOP-28, Pb-Free, Tape and Reel	Q1, Q2			1	ON Semiconductor	MMBT4242A,TIG	Volume Production	Samell	231360	0.0446	134.38
	SELECONBAT1B	Transistor Power MOSFET P-CH 20V 80A 8-Pin MOSFET Emboss T/R	Q3			1	Samell	SELECONBAT1B		Mouser	755-SELECONBAT1B	2.89	2.89
	10k	Chip Resistor, 10k Ohm, +/- 5%, 100 mW, -55 to 155 degC, 0803 (1608 Metric), RoHS, Tape and Reel	R1, R2			2	Vishay	CRCW08030K0FKEA	Volume Production				
	10kohm	RES SMD 10k OHM 1% 1/10W 0803	R3, R5			2	Vishay Bourns	MCT0603010020BP500	Volume Production	Mouser	594-MCT0603010020BP5	0.47	0.94
	1.8kohm	RES SMD 1.8K OHM 1% 1/10W 0803	R4			1	Yageo	RT0603RE073KBL	Volume Production	Arrow Electronics	RT0603RE073KBL	0.0799	0.0799
	3.24kohm	RES SMD 3.24K OHM 0.1% 1/5W 0803	R6			1							
	10kohm	RES SMD 10K OHM 1% 1/10W 0803	R7			1	Yageo	RT0603RE07120RL		Mouser	603-RT0603RE07120RL	0.11	0.11
	10k, 10k	RES SMD 10K OHM 1% 1/10W 0803	R8, R9, R10, R11, R12			5	Panasonic	E9A-34B110V	Volume Production	RSComponents	556890	0.508	2.53
	0.018ohm	Resistor	R13, R16			2	TE Connectivity	TLRP3A300R018PTE		Digi-Key	1712-TLRP3A300R018PTECT-ND	0.56	1.12
			R14, R15, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R42, R43, R44, R45, R46, R47, R48, R49, R50, R51, R52, R53, R54, R55, R56, R57, R58, R59, R60, R61, R62, R63, R64, R65, R66, R67, R72, R73, R74, R75, R76, R77, R78, R79, R80, R81, R82, R83, R84, R85, R86, R87, R88, R89, R90, R91, R92, R93, R94, R95, R96, R97, R98, R99, R100, R101, R102, R103, R104, R105, R106, R107, R108, R109, R110, R111, R112, R113, R114, R115, R116, R117, R118, R119, R120, R121, R122, R123, R124, R125, R126, R127, R128, R129, R130, R131, R132, R133, R134, R135, R136, R137, R138, R139, R140, R141, R142, R143, R144, R145, R146, R147, R148, R149, R150, R151, R152, R153, R154, R155, R156, R157, R158, R159, R160, R161, R162, R163, R164, R165, R166, R167, R168, R169, R170, R171, R172, R173, R174, R175, R176, R177, R178, R179, R180, R181, R182, R183, R184, R185, R186, R187, R188, R189, R190, R191, R192, R193, R194, R195, R196, R197, 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